Labor Market Sorting over the Business Cycle

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Abstract

I study whether workers selectively separate from their job matches in a recession. Using data from the Current Population Survey, I show that workers with short tenure and high wages are more likely to lose their jobs and become unemployed in recession periods. Moreover, the weekly history of workers from National Longitudinal Surveys suggests that workers having an experience of separation into unemployment for more than four weeks have higher separation probabilities later in their career. Moreover, the gap in separation probabilities between workers with and without separation experience becomes wider in recessions, which suggests an abundance of workers with small expected match surplus, or high separation rates, in the unemployment pool in bad times. I then test whether selective separation at the onset of the recession affects the job creation decision of employers. Since vacant jobs are more likely to be matched with unemployed workers, cyclical variation in the composition of unemployed workers changes the expected return to post vacancies. I introduce workers heterogeneous of outside values into the Diamond-Mortensen-Pissarides model with endogenous separation to see the macroeconomic implications of worker sorting through employment status over the business cycle. The key mechanism is that the firms tend to separate from their low surplus employees more frequently during recessions, entrants take this into account, and they reduce the number of vacancies created. Although the composition change amplifies cyclical fluctuations in job-finding rates and unemployment rates, its quantitative importance is small, compared to underlying forces to create cyclical fluctuations.

Keywords: cyclical unemployment, endogenous separation, labor market sorting, tenure dependence of separation probabilities, worker heterogeneity.
JEL codes: J63, E24, J64

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1 Motivation

A typical response of the labor market to a recession is characterized by a sharp increase in workers' separation probabilities at the onset of the recession and a more persistent reduction in job-finding probabilities of unemployed workers. If cyclical changes in labor market flows are different across *ex ante* heterogeneous workers, potential employers might adjust their job creation behavior to the composition of unemployed workers in a recession. In this project, I study how cyclical variation in the composition of unemployed workers is important in the job creation process over the business cycle. I argue that the fraction of unemployed workers that are low type increases in recession periods. Since entering firms have higher probabilities of matches to unemployed workers, this cyclical change in worker composition lowers the entry value.

To support this idea empirically, I show the unemployed worker pool has more inflows of high wage and short tenure workers in recessions, using the Current Population Survey (henceforth, CPS) micro-data. Workers in the lowest tertile wage group in an occupation and year cell have monthly job losing probabilities higher by 0.15 percentage points than the top tertile wage group. In recessions, the gap between wage groups disappears, implying an average unemployed worker has a higher wage in her previous job.

On the other hand, workers in the shortest 33 percent tenure group are 0.75 percentage points more likely to become job losers within a month, and the difference widens to 1 percentage point in recession periods. If the short tenure of workers implies a low match surplus, and this component stays persistent over worker career, having more short tenure in the match pool indicates having higher probabilities of small match surplus to the firms. I use the long panel of workers from National Longitudinal Surveys of Youth 1979 (hence, NLSY79) to test the persistence of the low match surplus component over workers' labor market transition. Having an experience of transition from employment to unemployment (EU transition) longer than four weeks increases average weekly EU transition rates by 0.2 percentage points, compared to workers without such experiences. During recessions, this effect amounts to 0.4 percentage points.

The data suggest that there exists a worker group with relatively higher wages and shorter tenure, and this worker group is more vulnerable to cyclical employment risks than others. A typical example of this kind of worker is lemons. When screening and signaling are imperfect in the hiring process, the firms that end up hiring low-type workers frequently separate from these workers. These firms are more likely to do so in recessions since the revenue of the match declines. If the wage cannot adjust freely to the decline in match productivity, the match ends with higher probabilities.

Another example of the worker group with short tenure and high wages is low productivity workers working in high-paying firms. If cross-sectional wage distribution within a firm is restricted so that the wage cannot reflect the productivity of workers entirely, firms separate from those workers more frequently, and the selection over time implies that the short-tenured worker is more likely to get fired.

I construct a simple Diamond-Mortensen-Pissarides (hence, DMP) type model to account for the cross-sectional difference in cyclical fluctuations in separation probabilities. Workers are heterogeneous in terms of the disutility of work, and their productivity. Low match surplus workers have high disutility of work and low productivity, and high match surplus ones have low disutility of work and high productivity. Endogenous separation happens when an overhead cost shock hits the existing match and when the realized cost is higher than the match surplus. As a result, at a given wage, only low-match surplus workers separate more frequently. Because of selection over the tenure, short labor market tenure is associated with higher probabilities of being low type and higher separation probabilities. Match-specific productivity implies a positive (negative) correlation between wage and the match surplus(separation probabilities).

In a recession, low-type workers with low match-specific productivity experience the largest increase in separation probabilities. These workers have a higher wage because of higher disutility from work, and they are more likely to be short-tenured, which replicates the data pattern observed from the CPS and the NLSY79.

The selective separation process means that the pool of unemployed workers changes over the business cycle. As long as the match is random across the type of workers, employers are more likely to match with low-type workers and have a smaller match surplus. Thus, the free entry condition implies that the labor market tightness decreases because of the composition effect. To quantify the importance of this margin, I compare the baseline impulse-response function to the counterfactual impulse-response function without composition effects in the free entry condition. The labor market sorting over the business cycle implies larger and more persistent dynamics in job-finding probabilities and unemployment rates, but it has limited quantitative importance due to the fast churning of the U.S. labor market.

2 Literature Review

This paper is in line with recent literature on the effect of worker heterogeneity on labor market flows. Using CPS microdata to observe the short career history of workers, Hall and Kudlyak (2019) assume five types of worker groups with different transition rates and decompose labor market flow rates as a mixture of heterogeneous exponential processes across long-term jobs, short-term jobs, non-participation, and unemployment. Morchio (2016) analyzes NLSY79 and shows that prime-age unemployment is concentrated for a segment of workers and unemployment spell and separation history among young workers is a strong predictor of staying unemployed and frequent separations at the prime age. He argues that information frictions on worker type and learning over the worker career are essential in explaining these facts. This paper introduces a cyclical component into this literature and shows that the workers with separation history are more vulnerable to cyclical shocks.

This paper extends the equilibrium model of labor market dynamics by introducing worker heterogeneity both in terms of hires and separations. While recent empirical studies em-

phasize the heterogeneity both in job-finding and separation probabilities, equilibrium unemployment models usually endogenize on the hiring process so that the former component depends on the worker type. These papers often base themselves on the empirical evidence from experimental episodes on duration dependence of callback rates (Oberholzer-Gee (2008), Kroft et al. (2013), Eriksson and Rooth (2014)); long-term unemployed are less likely to pass the initial process of hiring. When worker heterogeneity can be observed even imperfectly in the hiring process, longer unemployment duration signals the workers being low type (Doppelt (2016), Jarosch and Pilossoph (2019)). This paper, on the other hand, argues that the short job tenure signals the workers being low type and that the selective separation pattern in recession implies the pool of unemployed workers change in a less profitable direction than what a purely random separation suggests.

This paper is in line with the literature about information asymmetry in the labor market. Gibbons and Katz (1991) and Nakamura (2008) incorporate selection in separation into the hiring process. Even when worker type is unobservable, worker history is observable. Consequently, employers make a hiring decision based on the separation history of the worker. This paper assumes random matching. So, firms do not know which type of workers they meet, and *before the entry*, the free entry condition takes the composition of unemployed workers into account. This assumption simplifies the state variable of an individual worker and makes the model tractable. Moreover, this assumption makes the job-finding probability of workers, regardless of their types, decreases in recession by the same fraction. On the one side, Nakamura (2008) and Kroft et al. (2013) show the signaling of bad employment history has a weaker effect during recessions using U.S. data. On the other side, Biewen and Steffes (2010) report that there is no evidence of the negative impact of unemployment on job-finding being weaker during recessions in Germany. I rather take a neutral view and assume there is no selection in the hiring process at all.

This is paper aligns with recent findings of cyclical changes in the composition is unem-

ployed workers. Mueller (2017) also takes cyclical variation in the pool of unemployed workers into account and explores the impact of such change on job creation incentives by firms. He argues the unemployed worker pool in the CPS exhibits a larger fraction of workers with high wages from their previous jobs during recessions, and this comes from the high cyclicality of separation rates of those workers but not from job-finding sides. However, he concludes that this composition change dampens cyclical fluctuations in job-finding and unemployment rates, as workers with higher wages in their last job are more productive and potential employers benefit from having higher probabilities of being matched to them. This paper argues the opposite that the workers that additionally come into unemployment during recessions are more likely to give a lower value to the potential employers, because of their short tenure at their last job. Consequently, in this paper, the composition effect leads to the higher cyclicality of job-finding and unemployment rates. This paper emphasizes the role of worker heterogeneity as well as match-specific productivity in replicating the data pattern. Lise and Robin (2017) incorporate two-sided heterogeneity between workers and firms into the search-matching model with business cycle shocks. It builds a tractable framework to analyze the evolution of worker-firm match distribution over the business cycle. The key prediction of the model is that low-type workers are fired, especially by high productivity firms when the economy enters a recession. This paper reports empirical phenomena that are consistent with that prediction. The calibrated model in this paper delivers a similar result that a recession is characterized by workers with a high wage for reasons other than their productivity are more likely to flow into unemployment in bad times. The implication of such separation processes is lower job-finding rates appearing in the free entry condition and more sluggish recovery in the labor market after a negative shock.

3 Separation Probabilities over the Business Cycle

3.1 Cyclical Pattern of Separation Probabilities over Wage and Job Tenure Groups

To check test how cyclical firing risk accrues differently to workers across different wages and job tenures, I estimate firing probabilities over different wage and job tenure bins along with worker-level controls. For the wage bin, I use the Annual Social and Economic Supplement (ASEC) of the CPS. Then, using the short panel structure of the CPS, I merge the ASEC samples with the CPS core samples in the next month and estimate the probability that someone gets fired between the ASEC survey in March and the CPS core survey in April. The dependent variable $f_{i,t}$, where i is worker and t is the year, is defined as what follows.

$$f_{i,t} = \begin{cases} 1 & \text{if worker i is employed in ASEC sample} \\ & \text{and becomes job loser in the next month survey} \\ 0 & \text{if worker i is employed in ASEC sample} \\ & \text{and remains employed in the next month survey} \end{cases} \tag{1}$$

Then, I construct three wage bins, where each worker is assigned by the rank in each year and occupation cell, after excluding the top 5 and the bottom 5 percent of wage earners from my samples. I regress the following linear probability model.

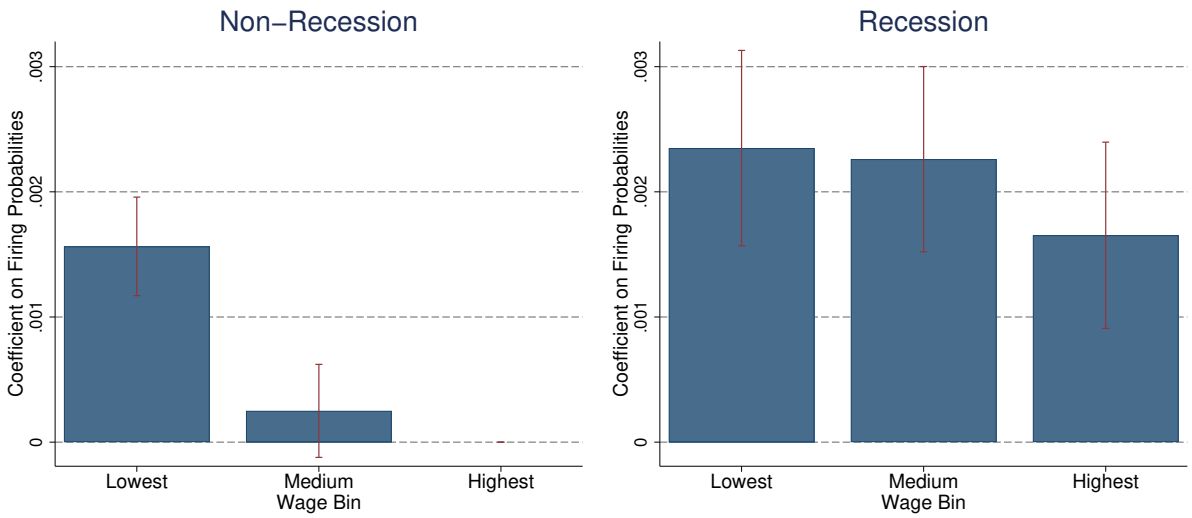
$$f_{i,t} = X'_{i,t}\beta + \sum_{q=1}^{3} \alpha_{q,normal} \mathbf{1}_{i,t}^{q,normal} + \sum_{q=1}^{3} \alpha_{q,recess} \mathbf{1}_{i,t}^{q,recess} + \nu_{i,t}. \tag{2}$$

In Equation 2, $\mathbf{1}_{i,t}^{q,normal}$ is a dummy that takes the value of one if the worker i belongs to the wage bin q and time t is non-recession periods. Likewise, $\mathbf{1}_{i,t}^{q,recess}$ is 1 if the worker i

belongs to the wage tertile bin q and the time t is in recessions. $X_{i,t}$ includes age, squared age, and dummies on race, occupation, industry, gender, and education level.

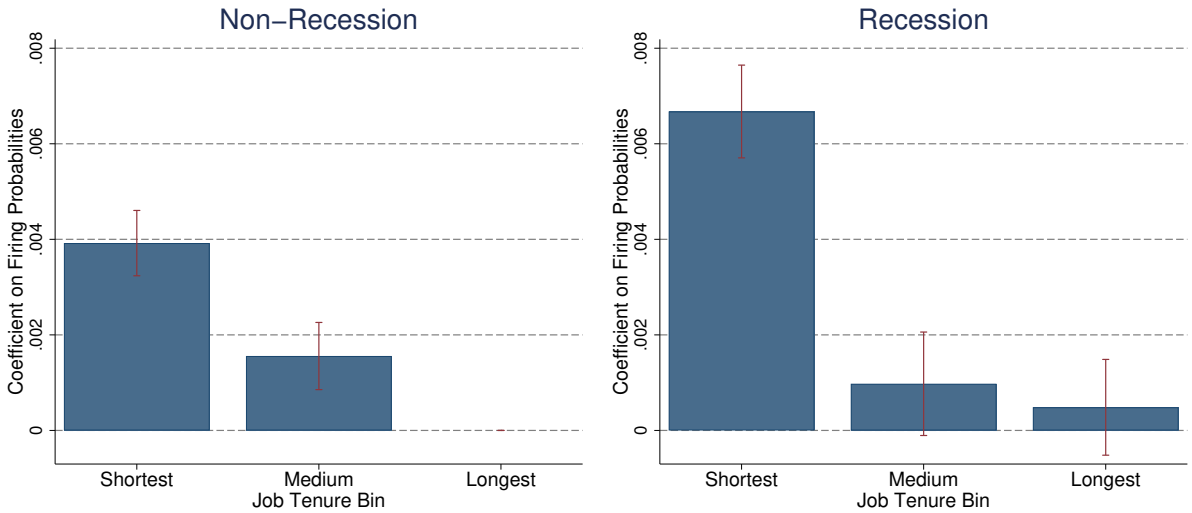
In order to estimate job-losing probabilities over job tenure bins, I use the Job Tenure and Occupational Mobility Supplement (JTOM) of the CPS. The firing dummy is defined similarly as before, and the tenure tertile bin is constructed by using workers' tenure rank in each occupation cell, regardless of the year. I exclude workers with job tenure shorter than one month and longer than twenty years. The controls are the same as the wage bin regression. I normalize coefficients on the highest wage bin and the longest job tenure bin in the non-recession periods to zero.

Figure 1: Job Losing Probabilities over Wage Groups



Data Source: IPUMS-CPS, ASEC, $N = 707,273$, $R^2 = 0.0075$.
The coefficient is normalized to the highest wage group in non-recession periods

Figure 2: Job Losing Probabilities over Tenure Groups



Data Source: IPUMS-CPS, JTOM, $N = 375,031$, $R^2 = 0.0126$.
The coefficient is normalized to the highest wage group in non-recession periods

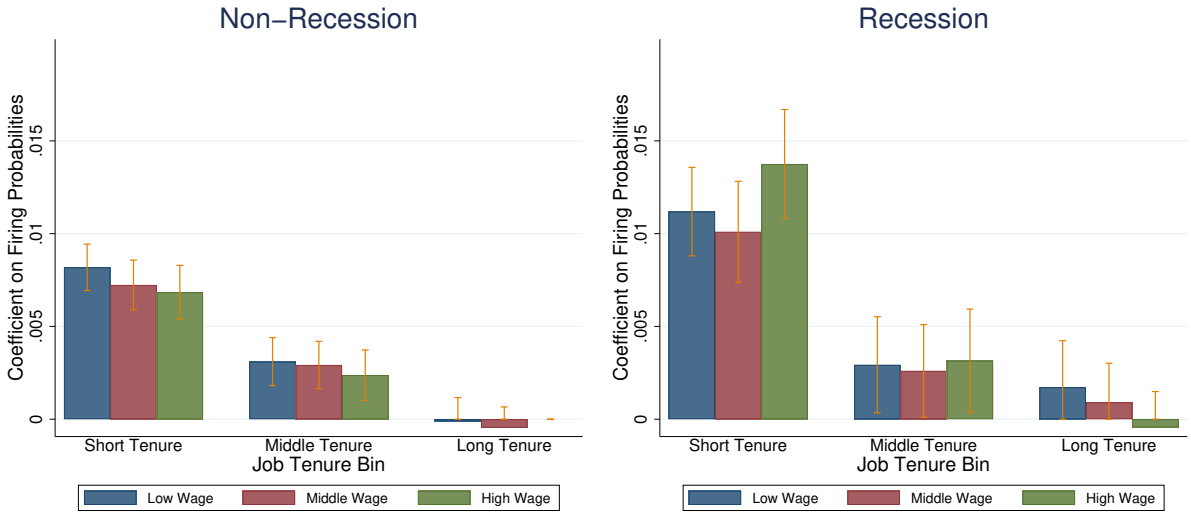
Figures 1 and 2 visualize the coefficient estimates. In non-recession periods, the workers in the first wage bin are 0.15 percentage points more likely to become job losers within a month, compared to the ones in the third wage bin. However, this difference is insignificant during a recession, and the highest wage group has a larger increase in separation rates than the lowest wage group, which is significant at the 10 percent significance level. The tendency is robust to controlling for a linear trend, the metropolitan status, the part-time status of each worker, and including job leavers, but the difference is not significant under some specifications. This result is consistent with the recent findings in Mueller (2017). He shows that workers with higher wages have larger cyclicality of separation rates, making those workers account for a disproportionately larger fraction of the unemployed worker pool in recession periods.

On the other hand, if workers are classified by their job tenure, workers in the first tenure bin have a 0.75 percentage points higher probability of becoming a job loser within a month. Moreover, it is the workers with the shortest job tenures that experience the

largest increase in the firing probabilities. Only this group has an increase in the job-losing probabilities statistically significant at the 10 percent level, and the difference in the cyclical increase is also significant at the 10 percent level. The tendency and the significance of the gap between cyclical increments are robust to controlling for a linear trend, the metropolitan status, the part-time status of each worker, and including job leavers.

To check whether cyclical increases in job-losing probabilities are largest for the shortest tenure even after controlling for their wages, I merge the ASEC sample to the JTOM sample. The ASEC samples are surveyed in March of every year, and The JTOM samples are surveyed in either January or February of a year. As a result, CPS rotation groups that are reported in the JTOM for the first (fifth), second (sixth), or third (seventh) times can also appear in ASEC. I limit the final sample for this exercise to the individuals who are employed both in the JTOM and the ASEC surveys. The regression takes the job-losing dummy from March to April in a given year as dependent variables and contains both wage and tenure bin dummies. I recalculate the bin from the final sample.

Figure 3: Job Losing Probabilities over Joint Groups



Data Source: IPUMS-CPS, ASEC and JTOM, $N = 199,066$, $R^2 = 0.0099$.
The coefficient is normalized to the highest wage and longest group in non-recession periods

The regression results are shown in Figure 3. Due to the small number of observations, all estimates yield large standard errors. Again, the recession does not harm the survival of medium- and long-tenured groups. Within the short-tenure groups, the high-wage group experiences the largest increase in job-losing probabilities in recessions. Only this group experiences an increase in the job-losing probability significant at the 10 percent level.

These exercises give several takeaways. First, workers with lower wages and shorter job tenure are overall more likely to become job losers, and short tenure is more strongly associated with higher separation probabilities than low wage as shown from the magnitude of the estimated coefficients. Second, workers with higher wages and short tenure have distinctively larger separation probabilities during recessions than in normal times. If the short tenure indicates worker component associated with a smaller profit to firms, despite the higher wage, differential EU rates over tenure bins imply lower entry value from the composition change among unemployed workers.

3.2 Persistent Impacts of Separation Experience on Workers

Even if separation is a selective process to sort bad matches, it does not lead to composition changes in the pool of unemployed workers if the worker heterogeneity does not persist over time. Answering this question requires a long panel of worker history. This section uses a long worker from NLSY79 to see if workers have different transition probabilities later depending on their work history[1]. The data set records a weekly history of each worker's employment status that started in 1979 until the 2010s. I test if the workers having EU transition before are more likely to separate from their jobs into unemployment or later, and if this pattern intensifies during recessions.

The EU separation variable is defined as follows. The worker is considered to separate

[1] The workers without separation history are entrants and re-entrants without any EU transition before.

from work only when the subsequent unemployment spell is longer than four weeks.

$$\tilde{f}_{i,t} = \begin{cases} 1 & \text{if worker } i \text{ is employed in week } t \\ & \text{and is unemployed from week } t+1 \text{ to } t+4 \\ 0 & \text{if worker } i \text{ is employed in week } t \\ & \text{and remains employed in any week from } t+1 \text{ to } t+4. \end{cases}$$

The four-week condition is to capture the idea that the worker may be involved in the job-to-job transition and have a short unemployment spell in between two jobs. However, most of the results are robust to removing the four-week condition and use only the employment status of one week after. By these definitions, the transitions from employment to non-participation are excluded from my samples. I also exclude samples that are not distinguishable between non-participation and unemployment.

The main regressor, EU separation experience dummy, is defined in the following equation.

$$\tilde{g}_{i,t} = \begin{cases} 1 & \text{if } max\{\tilde{f}_{i,s}\}_{s<t} = 1 \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

The worker has $g_{i,t} = 0$ if the worker separates for the first time in data at t. The regression specification is given as

$$\tilde{f}_{i,t} = X'_{i,t}\beta + \alpha_0 \tilde{g}_{i,t} + \alpha_1 \tilde{g}_{i,t} \mathbf{1}_t^{recess} + \nu_{i,t} \tag{4}$$

where $X_{i,t}$ contains controls including age, squared age, and dummies on education level, sex, occupation, industry, race, and recession. Then, α_1 can be interpreted as the additional effect of the recession on the separation probabilities of workers with separation

experience.

Table 1: NLSY79 Regression Estimates

Dependent Variable	Separation Probability (in %)		
	(1)	(2)	(3)
Separation Experience	0.246***	0.171***	0.245***
	(0.007)	(0.004)	(0.007)
Recession	0.047***	0.039***	0.046***
	(0.009)	(0.006)	(0.009)
Recession ×	0.264***	0.004	-0.024
Separation Experience	(0.026)	(0.013)	(0.053)
Hourly Wage			0.005***
			(0.000)
Hourly Wage × Recession ×			0.025***
Separation Experience			(0.004)
N	2640113	4902911	2231192
R-squared	0.001	0.001	0.001

Notes: ***,**,*: significant at 1, 5, 10 percent, respectively. Column (1) and (3) use young (age < 35) worker samples in NLSY79. Column (2) uses all age samples from 25 to 60. I use samples from 1998 because the occupation and industry codes are not collected before then.

Table 1 summarizes the estimation results. Having separation experience strongly predicts higher EU transition rates later, particularly for young workers. This finding is in line with the empirical findings in Morchio (2016). He shows that the separation experience of young workers leads to higher separation probabilities at their prime age. For the young worker samples, a worker having a separation experience before has a weekly separation probability higher by 0.25 percentage points than workers without any EU transition experience.

This tendency is even stronger in recession; having a separation experience is associated with a 0.31 percentage points total increase in separation probability in bad times, compared to only 0.05 percentage points increase for an average worker without such an experience. The estimates suggest the importance of separation experience on young workers relative to old workers in recession periods. The coefficient on the cyclical com-

ponent is smaller and less significant for full samples in NLSY79 as in column (2). When the regression includes hourly wage, and the additional effect of hourly wage on workers with separation experience, the level effect loses magnitude and significance, as shown in column (3).

To summarize, workers with a separation experience behave significantly differently later in the labor market. They are more likely to separate from their jobs, and cyclical shocks make them separate particularly more than workers without such experiences. As a result of increases in job separation probabilities heterogeneous across workers, there are more unemployed workers with multiple separation experiences in downturns, which the job creation side may take into account.

To see the implication of the worker sorting over the business cycles, I construct a frictional labor market model with worker heterogeneity and endogenous separation in the next section. The goal of the model is to generate a group of workers with high wages and short job tenure to account for the cyclical patterns in firing probabilities, and the model achieves this by introducing a component of worker type, disutility from working, increasing wage but decreasing the match surplus. The model delivers qualitatively similar results on the cyclical pattern on separation probabilities. Then, I compare the baseline impulse-response function to a counterfactual impulse-response function without worker sorting.

4 Model

I build a parsimonious frictional labor market model with worker heterogeneity to account for the cross-sectional patterns in separation probabilities; a worker with higher wage and shorter tenure is associated with the largest cyclical increase in separation probabilities. If the wage reflects a larger match surplus, part of which the firm perceives as profit, the overall tenure is longer for these workers through selection, contrary to the data.

Consequently, there has to be another wage-increasing factor that deteriorates the match

surplus. As described above, such workers can be thought of as lemons in the labor market or low-type workers working for high productivity firms. Firms happen to hire low productivity workers after an imperfect screening process and tend to separate from them more frequently. Alternatively, firms subject to wage equalization constraints, to some extent, choose to fire workers with the lowest productivity.

To generate a group of workers with high wages and a small match surplus, I assume workers with low productivity have high disutility of work. A high wage might result from both high disutility of work, in the case of low surplus workers, and high productivity, in the case of high surplus workers. At a given wage, the separation is concentrated around the low-type workers. Through selection, short tenure workers are more likely to be low type, and the separation probability is higher for short tenure workers. On the other hand, the cyclical shock hits mostly low-type workers with a higher wage, because the decrease in match surplus has a larger marginal effect on separation probabilities for low-type workers.

The model builds on the standard Diamond-Mortensen-Pissarides model (Pissarides (2000), Chapters 1 and 6) and additionally introduces worker heterogeneity and match-specific productivity. The economy consists of risk-neutral workers and firms. Each worker has a discrete type $\theta \sim m(\theta)$. The type affects both productivity and the disutility of work. One firm matches to one worker, after posting vacancies that incur costs to the firm. Once a firm matches with a worker, they draw match-specific productivity ξ from a time-invariant distribution, $H(\xi)$.

Each worker-firm match encounters an overhead cost shock with a Poisson rate of Ψ, which is potentially time-varying. Conditional on the shock, the overhead cost is distributed as continuous distribution $F(c)$ with full support over $\mathfrak{R}^+$. In order to continue the production, the match has to pay the overhead cost. In equilibrium with the possibility to negotiate in case of the match continues only if the realized cost is smaller than the match surplus, defined as the additional value of the match relative to the outside value of the worker.

To put this piece into the standard real business cycle framework, I assume that there is a representative household. It gathers all labor income together and decides how much to consume and to save. Their savings are used to produce by the firm-worker matches or to create vacancies. All the profits are pooled together and paid to the households, net of new vacancy costs. Assume an individual worker-firm match has the following decreasing return to scale production technology.

$$y = \xi\theta k^{\alpha},$$

where k is the capital input of individual match, with the rental rate r, ξ is firm-side productivity, and θ denotes worker type.
After maximization, a match earns the following revenue per period.

$$\pi(\theta, \xi) = (\theta\xi)^{\frac{1}{1-\alpha}} r^{\frac{-\alpha}{1-\alpha}} (\alpha^{\alpha} - \alpha^{\frac{1}{1-\alpha}}).$$

4.1 Workers

Workers are risk-neutral. Employed workers receive wages and separate from their jobs exogenously and endogenously. Working incurs disutility of work $b(\theta)$ to them, and the disutility is type-specific. The following equation describes the values of workers with type θ, match specific-productivity ξ, and wage w.

$$\begin{aligned} r\bar{E}(\theta, \xi, w|\Omega) = w - b(\theta) + (\chi + \Psi\mathbb{P}\left[S(\theta, \xi|\Omega) < c\right])(U(\theta|\Omega) - \bar{E}(\theta, \xi, w|\Omega)) \\ + \frac{d}{dt}\bar{E}(\theta, \xi, w|\Omega), \end{aligned}$$

where $S(\theta, \xi|\Omega) := \bar{V}(\theta, \xi, w|\Omega) + \bar{E}(\theta, \xi, w|\Omega) - U(\theta|\Omega)$ is the match surplus function.

Ω is the aggregate state variable, composed of the frequency of cost shock Ψ, common across agents, and the distribution of workers $g(\xi, \theta)$. I assume the agents can bargain how to split the overhead cost freely, so the match ends only when the cost exceeds the match surplus. Consequently, the match ends with the exogenous rate ξ and endogenous rate $\Psi\mathbb{P}\left[S(\theta, \xi|\Omega) < c\right]$, the probability that the overhead cost is larger than the match surplus multiplied by the Poisson rate. The worker is insured against the overhead cost shock. As a result, the firm pays all overhead costs if the match decides to pay the cost. The wage determination takes into this expected overhead cost. The wage is pinned down below through the Nash-bargaining.

When the worker is unemployed, the value can be expressed as the following equation.

$$rU(\theta|\Omega) = p(\tau) \int \big(E(\theta, \xi|\Omega) - U(\theta|\Omega)\big)\, dH(\xi) + \frac{d}{dt}U(\theta|\Omega)$$

In this equation, $p(\tau)$ is the rate at which each worker is matched to a firm. This rate is determined in equilibrium by the market tightness $\tau = v/u$. The flow utility of unemployed workers is normalized to zero.

4.2 Firms

The incumbent firm earns $\pi(\theta, \xi)$ from production every period and pays a wage to the worker. The firm value function is given as

$$\begin{aligned} r\bar{V}(\theta, \xi, w|\Omega) = \pi(\theta, \xi) - w + (\chi + \Psi\mathbb{P}\left[S(\theta, \xi|\Omega) < c\right])(-\bar{V}(\theta, \xi, w|\Omega)) \\ - \Psi\mathbb{E}[c|S(\theta, \xi|\Omega) \geq c]\mathbb{P}\left[S(\theta, \xi|\Omega) \geq c\right] \\ + \frac{d}{dt}\bar{V}(\theta, \xi, w|\Omega). \end{aligned}$$

The firm pays all the overhead costs if the match is hit by the overhead cost shock and

decides to pay the cost. The outside value for the firm is normalized to one.

4.3 Wage and Match Surplus

The wage is pinned down by the Nash-bargaining between the firm and the worker. To be specific, the worker and the employer bargain over wage to maximize the following object, with firm share parameter η.

$$w = \arg\max_{w} \left(\bar{E}(\theta, \xi, w|\Omega) - U(\theta|\Omega)\right)^{1-\eta} \left(\bar{V}(\theta, \xi, w|\Omega)\right)^{\eta}$$

Denote the resulting wage as $w(\theta, \xi|\Omega)$ and define $V(\theta, \xi|\Omega) := \bar{V}(\theta, \xi, w(\theta, \xi|\Omega)|\Omega)$ and $E(\theta, \xi|\Omega) := \bar{E}(\theta, \xi, w(\theta, \xi|\Omega)|\Omega)$. Then, the first order condition of the problem above implies

$$\begin{aligned} E(\theta, \xi|\Omega) - U(\theta|\Omega) &= (1-\eta)S(\theta, \xi|\Omega) \\ V(\theta, \xi|\Omega) &= \eta S(\theta, \xi|\Omega). \end{aligned}$$

The function $S(\theta, \xi|\Omega)$ satisfies the following equation.

$$\begin{aligned} rS(\theta, \xi|\Omega) = \pi(\xi|\Omega) &+ \left(\chi + \Psi\mathbb{P}[S(\theta, \xi|\Omega) < c]\right)\left(-S(\theta, \xi|\Omega)\right) \\ &- b(\theta) - p(\tau)\int \left(E(\theta, \tilde{\xi}|\Omega) - U(\theta|\Omega)\right) dH(\tilde{\xi}) \\ &- \Psi\mathbb{E}[c|S(\theta, \xi|\Omega) \geq c]\mathbb{P}[S(\theta, \xi|\Omega) \geq c] \\ &+ \frac{d}{dt}S(\theta, \xi|\Omega) \end{aligned}$$

As a result, it is sufficient to keep track of only $S(\theta, \xi|\Omega)$ to determine the flow of the labor market, not $E(\theta, \xi|\Omega)$ and $V(\theta, \xi|\Omega)$ separately. Unlike in the standard DMP model, negotiated wage has to take expected overhead costs into account. Thus, the firm agrees

to pay the overhead cost if the shock hits and cut the wage rate accordingly. As a result, an employed worker is insured against this overhead shock, and the firm gives only the wage net of the expected cost. The worker pays an η fraction of the expected cost per unit of time. Then, the wage becomes constant at a given (θ, η). Whether this overhead cost is paid continuously or only when the shock hits the match does not affect the behavior of agents. However, it matters how different wage groups are associated with separation probabilities in the model. Since wages are usually stable within a match, I assume the worker is insured against those cost-related separation shocks.

$$\begin{aligned}\tilde{w}(\theta, \xi) = (1-\eta)\pi(\theta, \xi) + \eta b(\theta) + \eta(1-\eta)p(\tau) \int S(\theta, \xi|\Omega) dH(\tilde{\xi}) \\ - \Psi \mathbb{E}[c|S(\theta, \xi|\Omega) \geq c]\mathbb{P}[S(\theta, \xi|\Omega) \geq c].\end{aligned}$$

In this equation, the last term is the insurance fee the worker pays in periods without overhead cost shocks.

From now on, I denote the endogenous separation rate as $\delta(S(\xi, \theta|\Omega)) = \Psi \mathbb{P}[S(\xi, \theta|\Omega) < c]$ and the expected cost the match chooses to pay conditional on the cost shock as $\mathbb{C}[S(\theta, \xi|\Omega)] = \mathbb{E}[X(c)|S(\theta, \xi|\Omega] \geq c]\mathbb{P}\left[S(\theta, \xi|\Omega) \geq c\right]$ for notational simplicity.

4.4 Matching Technology, Free entry, and the Market Clearing

The matching between firms and workers follows the Cobb-Douglas technology.

$$m(u, v) = Qv^{\gamma}u^{1-\gamma},$$

which gives $p(\tau) = Q\tau^{\gamma}$ and $\tau{=}v/u$.

The free entry condition is given as

$$c = q(\tau) \int \sum_{\theta} \frac{u(\theta)}{u} \eta S(\theta, \xi|\Omega) dH(\xi)$$

The distribution is characterized by the following equation.

$$\dot{g}(\theta, \xi) = u(\theta)p(\tau)h(\xi) - (\chi + \delta(S(\theta, \xi|\Omega)))g(\theta, \xi)$$

The representative household gathers all income from wages and firm profits. The household has log utility from consumption and discount rate ρ. The representative household decides how much to save and consume given the economy-wide interest rate r^C. Then, the standard Euler equation characterizes the optimal policy of the household.

$$\frac{d}{dt}\frac{1}{c(\Omega)} = (\rho - r^C(\Omega))\frac{1}{c(\Omega)},$$

which gives the interest rate as $r^C = \rho$ and the rental rate of capital as $r = \rho + \delta$ in the steady-state.

Lastly, the final good and the capital market clearing conditions are given as

$$\begin{aligned}
\frac{d}{dt}K(\Omega) + c(\Omega) + c_v v(\Omega) &= \left(\frac{\alpha}{r(\Omega)}\right)^{\frac{\alpha}{1-\alpha}} \sum_{\theta} \int (\xi\theta)^{\frac{1}{1-\alpha}} g(\theta, \xi) d\xi \\
&\quad - \sum_{\theta} \Psi \int \mathbb{C}(S(\theta, \xi|\Omega)) g(\xi, \theta) d\xi - \delta K(\Omega) \\
K(\Omega) &= \left(\frac{\alpha}{r(\Omega)}\right)^{\frac{1}{1-\alpha}} \sum_{\theta} \int (\xi\theta)^{\frac{1}{1-\alpha}} g(\theta, \xi) d\xi
\end{aligned}$$

By the Walras' law, the market-clearing conditions ensure that the budget constraint of

the representative housed holds with equality in equilibrium.

4.5 Equilibrium under the Aggregate Uncertainty

When aggregate fluctuations come from a stochastic process of the aggregate productivity, the fluctuations in the labor market are too small, well-known as the Shimer puzzle (Shimer, 2005) in the literature. To amplify cyclical responses of the labor market, suppose that the aggregate uncertainty comes from exogenous shocks to the overhead cost frequency Ψ, which follows the Ornstein-Uhlenbeck process.

$$d(\Psi_t - \Psi) = -\beta(\Psi_t - \Psi)dt + \sigma dW_t$$

where W_t is the Wiener process, β mean-reversion parameter, σ variance of the driving force. A positive shock to Ψ_t increases separation probabilities through more frequent overhead cost shock and smaller match surplus from an increase in expected overhead cost. Entry is also reduced by the decrease in the expected match surplus.

Following Reiter (2010) and Ahn et al. (2018), I discretize the space for ξ and linearize[2] the system of equations that characterize the equilibrium path with aggregate uncertainty around the deterministic steady-state and apply the Blanchard-Kahn to pin down the forward-looking variables, such as the value functions and the consumption level. The system of equations is fully characterized in appendix 4. From now on, The dependency on the aggregate states is summarized into the t subscript; e.g., $S(\xi, \theta|\Omega_t) = S_t(\xi, \theta)$.

5 Calibration

In the numerical exercise below, I assume two types of workers. The time is in months. I parameterize the overhead cost distribution F with a log-normal distribution of parameters (μ_c, σ_c) and the match-specific productivity distribution H with a log-normal

[2] I use automatic differentiation codes made by Ahn et al. (2018) for linearization.

distribution of parameters (μ_ξ, ζ). The parameters in Table 2 are set exogenously.

Parameter	Meaning	Value
r^C	Discount rate	0.004
δ	Depreciation rate	0.008
α	DRS parameter with respect to capital input	0.3
$p_U(\tau_U)$	UE rate for unemployed workers	0.45
χ	Exogenous EU rate	0.02
$m(\theta_{high})$	Mass of high surplus type	0.8
$b(\theta_{high})$	Disutility of work for high surplus type	0
Mean$[\theta]$	Mean type	1
Mean$[\xi]$	Mean productivity	1

Table 2: Exogeneously Given Parameters

I use 0.004 for the discount rate, r^C, so that the annual interest rate becomes around 5 percent per year. Standard values are used for the depreciation rate for production capital, $\delta = 0.008$, the decreasing return to capital parameter, $\alpha = 0.3$, and the average monthly UE rate, $p_U(\tau_U) = 0.45$. The disutility of work for the high type is normalized to zero, and the mean type and mean productivity are also normalized to one. I set the exogenous EU rate to 0.02 so that the third tenure group has only an exogenous separation probability of 0.02 per month in the steady-state. The mass of high surplus type, 0.8, comes from the observation in Hall and Kudlyak (2019) that the fraction of Non-employment and Unemployment types account for 24% of workers ever participating in the labor market for women and 13% for men in the data.

The parameters in Table 3 are calibrated to match the steady-state moments in Table 4, using the simulated method of moments. I also put some penalty on the mean separation probabilities for the high type so that high type workers have negligible probabilities of endogenous separation. I truncate the log-normal distribution for ξ at the first percentile of the distribution. I further restrict the equilibrium so that firms with the lowest ξ in the truncated distribution also find it profitable to hire both types of workers.

Parameter	Meaning	Value
Ψ	Steady state frequency of the overhead cost shock	0.59
μ_c	Mean of log-normal distribution for cost shock	0.1
σ_c	Standard deviation of log overhead cost	0.64
$b(\theta_{low})$	Disutility of work for low surplus type	1.26
ζ	Standard deviation of log match-specific productivity	0.07
η	Firm share of match surplus	0.71
θ_{high}	Productivity of high surplus type	1.01

Table 3: Calibrated Parameters: Steady-State

The equilibrium implies the steady-state frequency of the overhead cost shock is high. The shock hits once every two months. The standard deviation of the log overhead cost is large, so the separation probability is a smooth function of match surplus. The disutility of work for the low surplus type is large, around 53 percent of the average wage, while the productivity difference is small. As a result, most of the difference in the separation probability comes from the heterogeneity in the disutility of work. The standard deviation of log match-specific productivity is small, because of the equilibrium restriction that even firms with the lowest ξ hire all types of workers[3]. The firm share of the match surplus is larger than the previous literature because workers' outside option is higher than the homogeneous productivity case due to the option value of unemployment. As a result, the labor share becomes larger for a given η, and the firm share parameter must increase to match the labor share.

[3]This assumption is needed to use log-linearization methods to solve the equilibrium with aggregate shocks.

Moment	Value	Target
EU rate	0.024	0.025
EU rate for high type	0.00	0
EU difference between wage groups(3/1)	0.002	0.002
EU difference between wage groups(2/1)	0.001	0.001
EU difference between tenure groups(3/1)	0.008	0.008
EU difference between tenure groups(2/1)	0.005	0.006
Labor share	0.60	0.60
P90/P10 in wage distribution	1.10	2.80
P50/P10 in wage distribution	1.04	1.7

Table 4: Model and Target Moments: Steady-State

The calibrated model can match most of the targeted moments, except for the wage dispersion measure, because of the equilibrium restriction leading to the low value of ξ, and the weight on those moments smaller than others.

Figure 4 shows bar graphs of separation probabilities over wage and tenure bins from the calibrated model. The calibrated model replicates the steady-state pattern on separation rates across wage and tenure groups. Workers with low wages and short tenure are more likely to separate from their employers in a given period. Moreover, the degree of tenure-dependency is larger than the degree of wage-dependency; i.e., tenure is a stronger predictor of EU transition rates than wage.

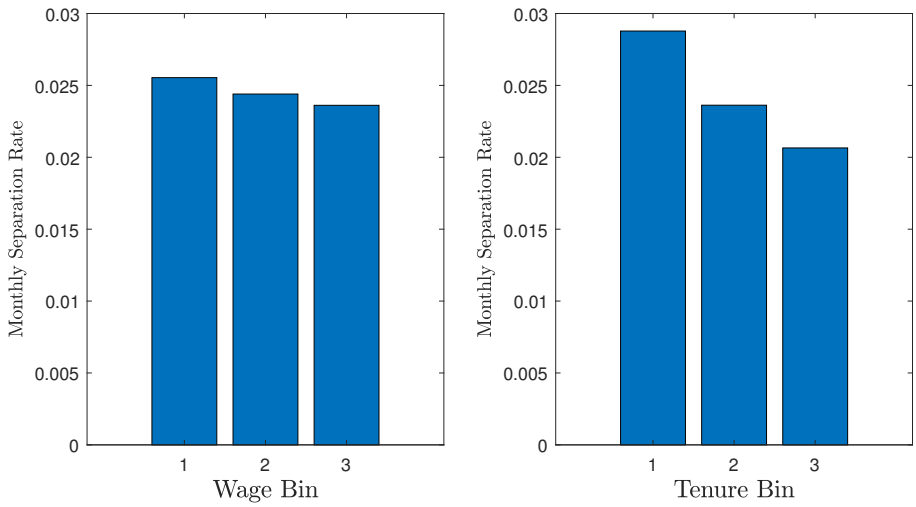


Figure 4: Steady-State Separation Probabilities over Wage and Tenure Groups

The left panel of Figure 5 expresses the density of workers over the match surplus across the two types. The low surplus type has smaller dispersion since the lower part of the distribution exits from employment at faster rates. Because of the selection of workers over time, the probability that a worker is a low surplus type decreases fast with the tenure, especially when a worker has a short tenure.

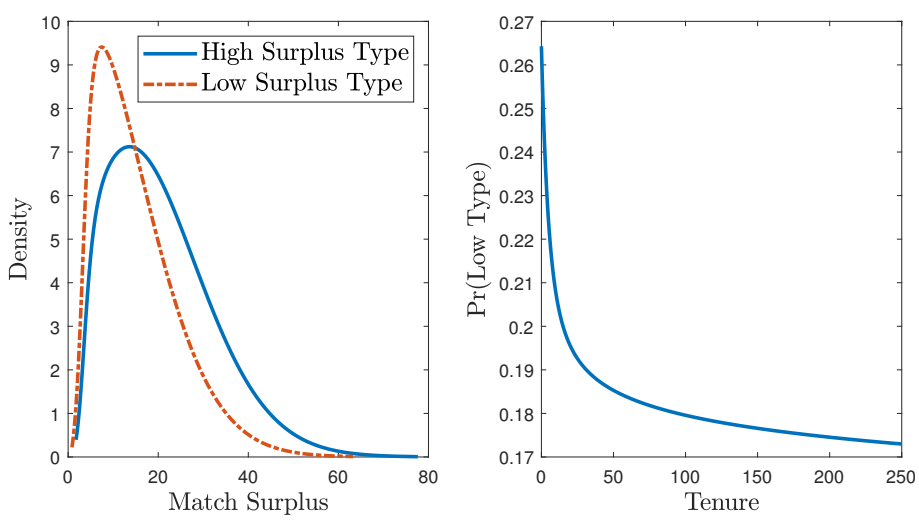


Figure 5: Steady-State Distribution of Workers

Parameters in the stochastic process for Ψ and the match elasticity, γ, also appear in the log-linearized system of equations characterizing the economy with aggregate uncertainty. The output Y_t is defined as the sum of individual output *before* paying the overhead cost. Tables 5 and 6 show the values of calibrated parameters and the targeted moments.

Parameter	Meaning	Value
β	Mean reversion parameter for Ψ process	0.118
σ	Volatility parameter for Ψ process	0.208
γ	Match elasticity	0.488

Table 5: Calibrated Parameters: Business Cycle

Moment	Value	Target
Std($\log(Y_t)$)	0.04	0.04
Std(UE_t)	0.05	0.05
Std(EU_t)	0.005	0.004
Autocorr(UE_t, UE_{t-3})	0.80	0.80

Table 6: Model and Target Moments: Business Cycle

6 Aggregate Implications

6.1 Impulse Response Functions

Figure 6 displays the impulse responses of labor market flow rates to one standard deviation Dirac-Delta shock to Ψ_t at $t = 0$, which decays at a constant rate afterward. An increase in the frequency of overhead cost shocks reduces the match surplus for all matches. Consequently, the job-finding rate decreases from 45 to 4.32 percent and recovers slowly. The separation rate jumps up by 0.5 percentage points per month and returns to the steady-state level at a faster speed.

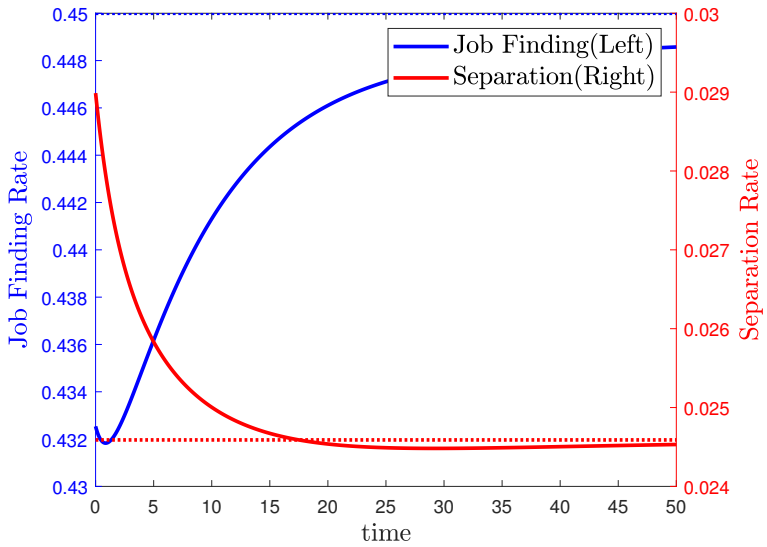


Figure 6: Impulse Responses Functions: Job-Finding and Separation Rates

Responses of aggregate variables are shown in Figure 7. In response to a shock raising the frequency of overhead cost shock by 20 percent, the unemployment rate reaches 5.7 percent, 0.5 percentage points higher than the steady-state level. The decrease in employment reduces output initially, and a long-run decrease in output comes from a decrease in production capital and interest rate higher than the steady-state level.

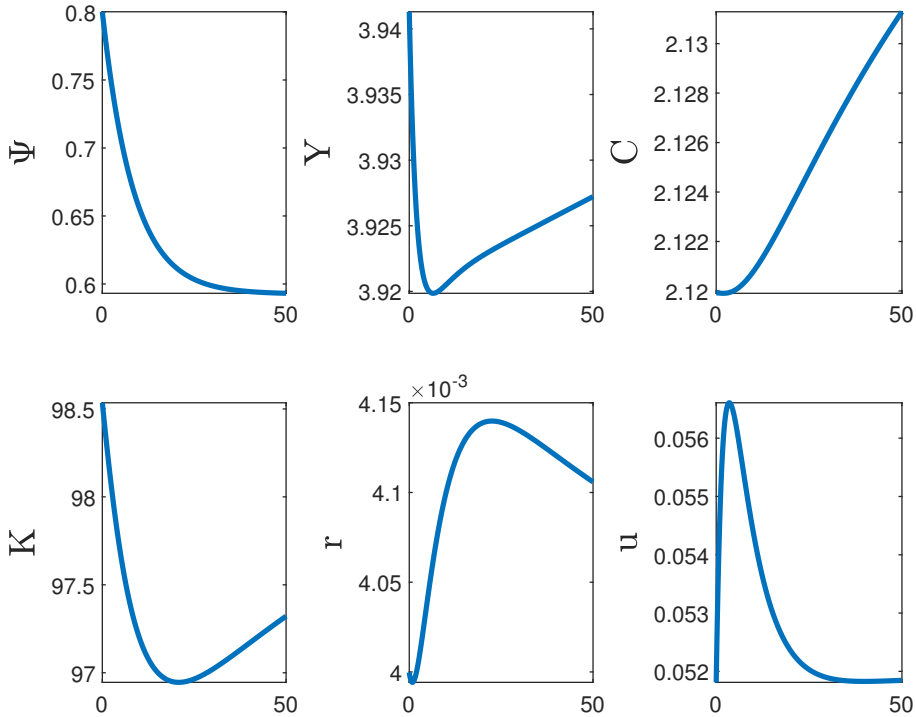


Figure 7: Impulse Response Functions: Aggregate Variables

6.2 Cyclical Increases in Separation Probabilities

Figure 8 shows increases in separation rates over different wage and tenure bins, instantaneously after the Dirac-Delta shock. The calibrated model gives patterns qualitatively similar to the ones in the CPS data; the last wage bin and the first tenure bin experience the largest increases in separation rates.

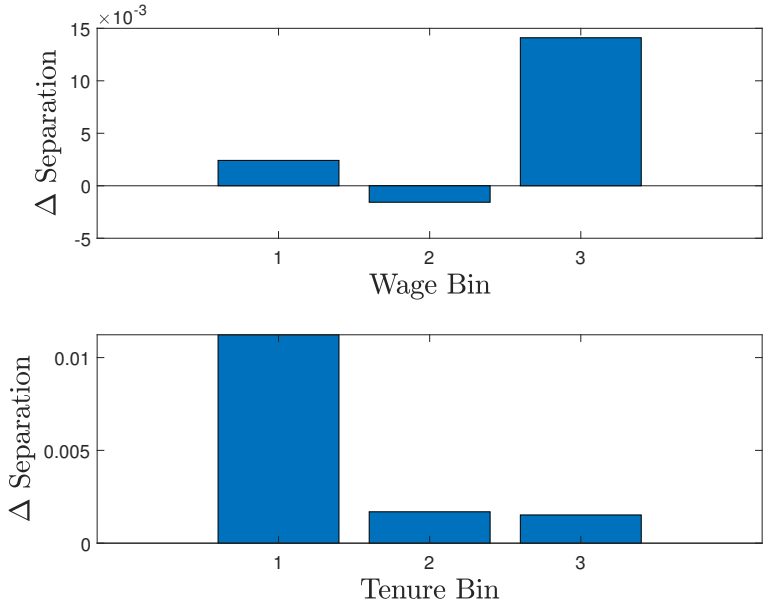


Figure 8: Increase in Separation Rates over Wage and Tenure Bins

A component that raises the wage but *reduces* the match surplus, the disutility of work in this model, is essential to break the monotone correspondence between tenure (or persistently low separation rate) and wage. Low-type workers with low match-specific productivity have higher wages because their disutility of work is higher, not because they are more productive. They are more likely to separate into unemployment because of this higher disutility of work. Having a short job tenure signals the type of worker through selection, so job tenure is a strong predictor of separation probabilities.

The cyclical shock in the frequency of overhead cost shock, Ψ_t, hits the low type workers with low match-specific productivity hardest, as the density of the overhead cost shock is higher around the value of the match with these workers. These workers have high wages and short job tenure, replicating the empirical data pattern. These workers can be thought of as the lemons that survived the screening process but gives a small profit to the employer.

6.3 Sorting and Persistence of Labor Market Variables

Figure 9 expresses changes in the composition of unemployed workers after the Dirac-Delta shock. The fraction of low-type workers in unemployment increases from 26.5 to 28 percent and stays higher than the steady-state level for a while. The change in the share of low type is modest in magnitude because separation probabilities recover quickly to the pre-recession level. Only a small fraction of matches with low-type workers are affected by the cyclical shock.

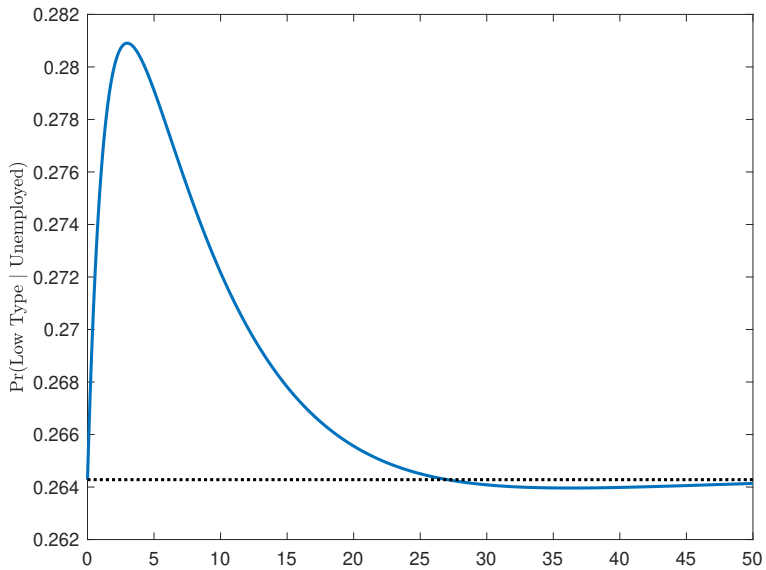


Figure 9: Fraction of Unemployment Workers that are Low Type

The selection of low type into unemployment implies the composition effect works to decrease the entrants' value from job creation in recession. To see the qualitative and quantitative implications of worker sorting over the business cycle, I compare the impulse responses of job-finding and unemployment rates from the benchmark model to the rates from a counterfactual scenario without worker composition effect in the job creation decision. I substitute the free entry condition of the benchmark model with the following equation.

$$c = q(\tau_t) \int \sum_{\theta} \frac{u(\theta)}{u} \eta S_t(\theta, \xi) dH(\xi).$$

The only difference is that the composition term, $u(\theta)/u$, uses the steady-state value instead of the actual composition term, $u_t(\theta)/u_t$. This is a partial equilibrium analysis, in the sense that the interest rate r_t as well as the value $S_t(\theta, \xi)$ determining separation rates between workers and employers stay the same as the benchmark case with sorting, and only the free entry condition determining labor market tightness changes.

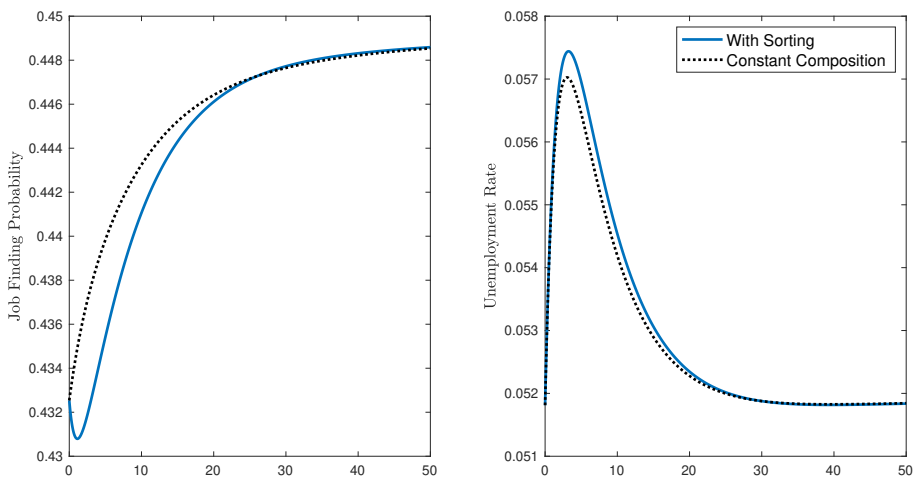


Figure 10: Impulse Responses with and without Worker Sorting

Figure 10 compares the responses of job-finding probabilities and the unemployment rates in the benchmark case and to the case without worker sorting. Sorting contributes to hump-shaped responses in job-finding rates since the composition among unemployed workers is a slow-moving state variable. Job-finding rates fluctuate more persistently, and the unemployment rate increases by more. However, the impact of sorting is quantitatively small, especially on unemployment rates.

The reason for the small quantitative impact is the failure of the model to generate a sufficient influx to unemployment through separation. Most of the initial increase in the

unemployment rate is due to the sharp decrease in job-finding probabilities. The model assumes no selection for the hiring process, so the small separation translates into a small impact of sorting. Data suggests that a higher amount of separation contributes to a substantial decrease in employment at the onset of the recession.

The impact of sorting can be magnified if the considered scenario is such that the Ψ stays high for a while, then goes back to the steady-state level, instead of the rapidly decaying Dirac-Delta shock. In this case, the separation may also play a large role in increasing the unemployment rate, with a higher amount of low-type workers in the unemployment pool[4].

7 Concluding Remarks

I interpret separation as a selective process for firms to sort low-type workers into unemployment. Data suggests the selection is stronger in recessions, as implied by increases in separation rates concentrated around workers with short tenure. Short tenure signals low-type workers in endogenous separation models. From a potential employers' perspective, stronger sorting in recessions means a lower value of creating new jobs, as they are more likely to end up hiring low-type workers. In spite of modest magnitude, sorting amplifies the impact of aggregate productivity shocks in the labor market through larger fluctuations in job-finding rates. This mechanism can provide recent literature on the regional variation of employment dynamics with explanations about why workers exposed to regional labor market shocks during the Great Recession still have lower employment rates even in the mid-2010s, as pointed out by Yagan (2019) and why massive local layoffs have negative spillover impacts on other unemployed workers at the timing of the event in Germany, as in Gathmann et al. (2017).

[4]Another way to amplify the role of sorting is to incorporate the directed search in the hiring process. If only some of the unemployed workers are mixed with the separated workers, due to the signaling effect of unemployment duration, for instance, even small amount of endogenous separation applied to low-type workers may change the entry value through composition effect by a larger amount.

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A Appendix

A.1 Cyclical Changes in Job-Finding Probabilities across Different Wage and Tenure Groups

This section shows that job-finding probabilities do not change systematically differently across wage and tenure groups over the business cycle. Define the job-finding dummy from the CPS questions in the following way similar to Equation 1.

$$h_{i,t} = \begin{cases} 1 & \text{if worker i is fired one month after the ASEC survey} \\ & \text{and becomes employed in the next or the second next month survey} \\ 0 & \text{if worker i is fired one month after the ASEC survey} \\ & \text{and remains unemployed in the next or the second next month survey} \end{cases}$$

As a result, the samples consist only of workers that lost their jobs after the ASEC and the JTOM survey. Their wage and tenure bin dummies measured in the ASEC and JTOM surveys are used as regressors. Other than the dependent variables and the bin regressors, the same controls as in Equation 2 are used in the OLS estimation.

Dependent Variable	Separation	Job-Finding Probability (in %)	
	Probabiltiy (in %)	Within 1 Month	Within 2 Months
Non-Recession			
Low Wage	0.20***	-2.22	-1.33
	(0.02)	(1.75)	(1.20)
Middle Wage	0.05**	-2.23	-2.30*
	(0.02)	(1.81)	(1.24)
Recession			
Low Wage	0.32***	-8.27***	0.67
	(0.05)	(2.99)	(1.98)
Middle Wage	0.25***	0.23	-3.29
	(0.05)	(3.13)	(2.16)
High Wage	0.21***	-5.55*	-1.04
	(0.05)	(3.27)	(2.18)
N	788,961	5,084	3,643

Table 7: Job-Finding and Separation Probabilities Across Wage Groups

Dependent Variable	Separation	Job-Finding Probability (in %)	
	Probabiltiy (in %)	Within 1 Month	Within 2 Months
Non-Recession			
Low Tenure	0.64***	0.44	-4.92*
	(0.05)	(0.66)	(2.77)
Middle Tenure	0.26***	0.32	-6.07**
	(0.05)	(0.70)	(2.94)
Recession			
Short Tenure	1.08***	-1.00	-2.35
	(0.07)	(0.80)	(3.34)
Middle Tenure	0.31***	-1.02	-7.40
	(0.07)	(0.96)	(4.02)
Long Tenure	0.08	-0.77	-4.78
	(0.06)	(1.01)	(4.21)
N	270,670	2,043	1,934

Table 8: Job-Finding and Separation Probabilities Across Tenure Groups

Tables 7 and 8 summarize the regression results. Due to the sampling restrictions, the number of observations is small for the job-finding probability regression. As a result, it is hard to implement a statistical test for the differences in cyclical decreases of job-finding probabilities across wage and tenure groups. Still, given that the monthly job-finding probability is around 43-45%, regression estimates feature a uniform decrease in job-finding probability during recession periods.

For the NLSY samples, the job-finding dummies are also defined similarly as Equation 3 but without the four-week restrictions. The regression specification is the same as Equation 4, and the samples for this regression are again restricted to workers younger than 35. Moreover, workers who are looking for a job after their first separation experience are excluded because their hiring dummies are mechanically correlated with the separation experience dummy.

$$\tilde{h}_{i,t} = \begin{cases} 1 & \text{if worker } i \text{ is unemployed in week } t \\ & \text{and is employed in week } t+1 \\ 0 & \text{if worker } i \text{ is unemployed in week } t \\ & \text{and remains unemployed in week } t+1. \end{cases}$$

Dependent Variable	Separation Probability (in %)		Job-Finding Probability (in %)	
	(1)	(2)	(3)	(4)
Separation Experience	0.22***	0.22***	-1.81***	-1.81***
	(0.01)	(0.01)	(0.26)	(0.26)
Hourly Wage	0.00***	0.00***	-0.25***	-0.25***
	(0.00)	(0.00)	(0.02)	(0.02)
Recession	0.07***	0.07***	-0.05	-0.05
	(0.01)	(0.01)	(0.41)	(0.41)
Recession ×	0.25***	-0.09	-0.69	-2.16
Separation Experience	(0.03)	(0.07)	(0.75)	(1.34)
Hourly Wage × Recession ×		0.04***		0.14
Separation Experience		(0.01)		(0.11)
N	2,229,259	2,229,259	62,461	62,461

Table 9: Job-Finding and Separation Probabilities and the Separation Experience

Table 9 shows the regression results. Workers with a separation experience are less likely to find a job in a week than labor market entrants. However, unlike separation probability results, these workers do not have a larger or smaller decrease in their job-finding probabilities during recessions. Likewise, workers with a higher hourly wage at their last jobs have a lower probability of finding a job in a given week, but their job-finding probabilities do not decrease by more significantly in a recession. Moreover, the recession dummy yields an insignificant and small coefficient estimate after controlling for separation experience and hourly wage. This estimate suggests that the lower job-finding probability during bad times may result from changes in the composition of unemployed workers. In recession times, more workers have higher hourly wages at their last jobs and separation experiences. As a result, the average job-finding probability of these workers is diminished.

A.2 Derivation of Nash Bargained Wage

Wage is set to maximize the following object.

$$w = \arg\max_{w} \left(\bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)\right)^{1-\eta} \left(\bar{V}(\theta,\xi,w|\Omega)\right)^{\eta}$$

Let's define $\bar{S}(\theta,\xi,w|\Omega) = \bar{V}(\theta,\xi,w|\Omega) + \bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)$. Then,

$$\begin{aligned} r\bar{S}(\theta,\xi,w|\Omega) = \pi(\xi|\Omega) &+ \left(\chi + \Psi\mathbb{P}[\bar{S}(\theta,\xi,w|\Omega) < c]\right)\left(-\bar{S}(\theta,\xi,w|\Omega)\right) \\ &- b(\theta) - p(\tau)\int \Big(E(\theta,\tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\, dH(\tilde{\xi}) \\ &- \Psi\mathbb{E}[c|\bar{S}(\theta,\xi,w|\Omega) \geq c]\mathbb{P}[\bar{S}(\theta,\xi,w|\Omega) \geq c] + \frac{d}{dt}S(\theta,\xi|\Omega) \end{aligned}$$

This gives

$$\frac{d\bar{S}(\theta,\xi,w|\Omega)}{dw} = 0,$$

and $\bar{S}(\theta,\xi,w|\Omega) = S(\theta,\xi|\Omega)$. Moreover,

$$\begin{aligned} \Big(r + \chi + \Psi\mathbb{P}\left[S(\theta,\xi|\Omega) < c\right]\Big)\frac{d\bar{E}(\theta,\xi,w|\Omega)}{dw} &= 1 + \Psi\eta f(S(\theta,\xi|\Omega))S(\theta,\xi|\Omega)\frac{dS(\theta,\xi|\Omega)}{dw} \\ &= 1 \end{aligned}$$

Likewise,

$$\Big(r + \chi + \Psi\mathbb{P}\left[S(\theta,\xi|\Omega) < c\right]\Big)\frac{d\bar{V}(\theta,\xi,w|\Omega)}{dw} = -1$$

The first order condition of the wage setting problem is

$$(1-\eta)\left(\bar{V}(\theta,\xi,w|\Omega)\right) = \eta\left(\bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)\right)$$

$$(1-\eta)\Bigg\{\pi(\xi|\Omega) - w - \Psi\eta\mathbb{C}(S(\theta,\xi|\Omega)) = \eta\Bigg\{w - b(\theta) - p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\,dH(\tilde{\xi})$$

$$+\frac{d}{dt}\bar{V}(\theta,\xi,w|\Omega)\Bigg\} \quad - \Psi(1-\eta)\mathbb{C}(S(\theta,\xi|\Omega)) + \frac{d}{dt}[\bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)]\Bigg\},$$

where $\mathbb{C}(S(\theta,\xi|\Omega)) = \mathbb{E}[c|c \leq S(\theta,\xi|\Omega)]\mathbb{P}[c \leq S(\theta,\xi|\Omega)]$.
Rearranging the equation above, using $(1-\eta)\frac{d}{dt}\bar{V}(\theta,\xi,w|\Omega) = \eta\frac{d}{dt}[\bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)]$,

$$\begin{aligned} w(\theta,\xi) &= (1-\eta)\pi(\xi|\Omega) - \eta b(\theta) + \eta p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\,dH(\tilde{\xi}) \\ &= (1-\eta)\pi(\xi|\Omega) - \eta b(\theta) + \eta(1-\eta)p(\tau)\int S(\xi,\theta|\Omega)dH(\tilde{\xi}). \end{aligned}$$

This expression gives

$$\begin{aligned} \big(r+\chi+\delta(S(\theta,\xi|\Omega))\big)\left(E(\theta,\xi|\Omega) - U(\theta|\Omega)\right) &= (1-\eta)\Bigg(\pi(\xi|\Omega) \\ &\quad - b(\theta) - p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\,dH(\tilde{\xi}) \\ &\quad - \Psi\mathbb{C}(S(\theta,\xi|\Omega)) + \frac{d}{dt}[E(\theta,\xi|\Omega) - U(\theta|\Omega)]\Bigg) \\ \big(r+\chi+\delta(S(\xi,\theta|\Omega))\big)V(\theta,\xi|\Omega) &= \eta\Bigg(\pi(\xi|\Omega) - b(\theta) - p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\,dH(\tilde{\xi}) \\ &\quad - \Psi\mathbb{C}(S(\theta,\xi|\Omega)) + \frac{d}{dt}V(\theta,\xi)\Bigg). \end{aligned}$$

Then,

$$\begin{aligned} E(\theta,\xi|\Omega) - U(\theta|\Omega) &= (1-\eta)S(\theta,\xi|\Omega) \\ V(\theta,\xi|\Omega) &= \eta S(\theta,\xi|\Omega), \end{aligned}$$

where again

$$rS(\theta,\xi|\Omega) = \pi(\xi|\Omega) - b(\theta) - p(\tau)\int(1-\eta)S(\theta,\tilde{\xi}|\Omega)dH(\tilde{\xi})$$
$$- \Psi\mathbb{C}(S(\theta,\xi|\Omega)) - (\chi + \delta(S(\theta,\xi|\Omega)))S(\theta,\xi|\Omega) + \frac{d}{dt}S(\theta,\xi|\Omega).$$

A.3 Full Characterization of the Economy with Aggregate Uncertainty

For notational simplicity, I summarize the dependency on the aggregate states to t subscript; i.e. $S(\xi,\theta|\Omega_t) = S_t(\xi,\theta)$, and other variables are denoted similarly. The full system of equations that characterize the stochastic equilibrium is described in the following equations.

$$\frac{d}{dt}g_t(\theta,\xi) = p(\tau_t)u_t(\theta)h(\xi) - (\chi + \delta_t(\theta,\xi))g_t(\theta,\xi)$$
$$\frac{d}{dt}K_t + c_t + c_v v_t = \left(\frac{\alpha}{r_t}\right)^{\frac{\alpha}{1-\alpha}}\sum_\theta\int(\xi\theta)^{\frac{1}{1-\alpha}}g_t(\theta,\xi)d\xi - \sum_\theta\Psi_t\int\mathbb{C}(S_t(\theta,\xi))g_t(\xi,\theta)d\xi - \delta K_t$$
$$K_t = \left(\frac{\alpha}{r_t}\right)^{\frac{1}{1-\alpha}}\sum_\theta\int(\xi\theta)^{\frac{1}{1-\alpha}}g_t(\theta,\xi)d\xi$$
$$d(\Psi_t - \Psi) = -\beta(\Psi_t - \Psi)dt + \sigma dW_t$$
$$\frac{d}{dt}S_t(\theta,\xi) = (r_t + \chi + \delta_t(\theta,\xi))S_t(\theta,\xi) - \pi_t(\theta,\xi) - b(\theta) - \Psi_t\mathbb{C}(S_t(\theta,\xi))$$
$$- p(\tau_t)\eta(1-\eta)\int S_t(\theta,\tilde{\xi})dH(\tilde{\xi})$$
$$\delta_t(\theta,\xi) = \Psi_t(1 - F(S_t(\theta,\xi)))$$
$$\frac{d}{dt}c_t^{-1} = (\rho + \delta - r_t)c_t^{-1}$$
$$c = q(\tau_t)\int\sum_\theta\frac{u_t(\theta)}{u_t}\eta S_t(\theta,\xi)dH(\xi)$$